JNTUH USED 24-11-17PM Set 2

Code No: 117EE **R13**

JAWAHARLAL NEHRU TECHNOLOGICAL UNIVERSITY HYDERABAD
B. Tech IV Year I Semester Examinations, November/December - 2017
LINUX PROGRAMMING
(Computer Science and Engineering)

Time: 3 Hours **Max. Marks: 75**

Note: This question paper contains two parts A and B.
Part A is compulsory which carries 25 marks. Answer all questions in Part A. Part B consists of 5 Units. Answer any one full question from each unit. Each question carries 10 marks and may have a, b, c as sub questions.

PART- A

(25 Marks)

1.a)	What are shell responsibilities?	[2]
b)	What are the applications of awk?	[3]
c)	What are hard links?	[2]
d)	Write about file locking?	[3]
e)	What are reliable signals?	[2]
f)	Differentiate threads and processes.	[3]
g)	What is IPC?	[2]
h)	Explain popen.	[3]
i)	What are Berkeley sockets?	[2]
j)	List the APIs for shared memory.	[3]

PART-B

(50 Marks)

2.a) Explain associative arrays.
b) Write a shell script to find the factorial of a number. [5+5]

OR

3.a) Develop an AWK program to summarize from the list of all processes, a count of processes run by every user (including root).
b) Write about text processing utilities. [5+5]

4. Differentiate between the following terms:
a) getc( ) Vs fgetc( ) b) stat( ) Vs fsat( )
c) printf( ) Vs fprint( ) d) scanf( ) Vs fscanf( ). [10]

OR

5.a) Explain the following system calls:
i) open( ) ii) seek( ) iii) read( ) iv) link( )
b) Explain directory handling system calls. [5+5]

6.a) Differentiate between fork( ) and vfork( ).

b) Write the syntax of six versions of exec functions and also explain how these functions differ from each other. [5+5]

OR

7. Write a c program that accepts two small numbers as arguments and then sums the two numbers in a child process. The sum should be returned by child to the parent as its exit status and the parent should print the sum? [10]

8. Write a program and explain how to transfer a large amount of data between two processes using Message queues. [10]

OR

9. Explain the following concepts about pipes:
   a) Pipes between two process
   b) Pipes among three process in a shell. [5+5]

10. Explain with a program how to copy file data from server to client using shared memory. [10]

OR

11.a) Explain briefly about the following socket APIs with clear syntax:
   i) socket( ) ii) bind( ) iii) listen( ) iv) accept( ) v) connect( )

b) Compare various IPC mechanisms. [5+5]

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